UNIVERSITY OF ENGINEERING AND TECHNOLOGY,PESHAWAR

ABBOTTABAD CAMPUS

Patient monitoring system

Submitted By:

Student Name	Registration Number	Roll No.
Amina Bibi	24ABELT0988	01
Iqra Noor	24ABELT1001	14
Maryam Shahzad	24ABELT103	16

Submitted To:

Sir Wajid

Course Title:

Electronic CIRCUIT 2

Department:

Electronic engineering

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PATIENT MONITORING SYSTEM

MODULE 1: TEMPERATURE MEASURING SYSTEM

1.TABLE OF CONTENTS:

2. Hardware Components Table

Item No.	Hardware Component	Physical Package / Model	Specifications	Primary Function in Hardware
1	Op-Amp IC 1	LM358 / TL072 (DIP-8 IC)	Dual Low-Power Operational Amplifier	Implements Stage 1 (Amplifier) and Stage 3 (Comparator).
2	Op-Amp IC 2	OP07 / LM741 (DIP-8 IC)	High-Precision Operational Amplifier	Implements Stage 4 (Relaxation Oscillator).
3	Temperature Module	NTC Thermistor Board	Active 3-Pin Sensor Module	Converts physical temperature into an analog voltage signal.

Item No.	Hardware Component	Physical Package / Model	Specifications	Primary Function in Hardware
4	Bench Power Supply 1	BK Precision 1670A	Regulated Dual DC Supply (±12 V)	Provides the main power supply for the operational amplifier stages.
5	Bench Power Supply 2	UNI-T UTP3315TFL	Regulated Variable DC Supply (0–37 V)	Used for calibration and simulated sensor input voltage.
6	Visual Indicator	Red LED (5 mm)	Forward Voltage ≈ 2 V	Indicates when the measured temperature exceeds the preset threshold.
7	Audible Indicator	Piezoelectric Buzzer	Operating Voltage: 5–12 V	Produces an audible alarm driven by the oscillator circuit.
8	Film Resistors	Through-Hole, ¼ W	1 Ω, 10 Ω, 99 Ω, 100 Ω, 470 Ω	Used for gain setting, filtering, hysteresis, and timing networks.
9	Ceramic Capacitors	Disc / Multilayer (Radial)	10 nF (103), 0.1 µF (104)	Implements low-pass filtering, noise suppression, and timing functions.
10	Probing & Jumpers	Banana-to-Alligator Clips / Jumper Wires	Standard Breadboard Accessories	Used for signal routing, power connections, and oscilloscope probing.
11	Prototyping Board	Solderless Breadboard	830 Tie-Points	Provides a platform for circuit assembly and hardware testing.

Challenges Faced During Hardware Procurement

A significant challenge during the hardware implementation phase was the procurement of the required electronic components. Many specialized components, including precision operational amplifiers, thermistor modules, and specific resistor and capacitor values, were not readily available in the local electronics markets of Abbottabad. As a result, considerable time was spent visiting multiple suppliers and searching for compatible components that met the project specifications while remaining cost-effective. Since some of the initially planned operational amplifiers were either unavailable or comparatively expensive, **LM741** and **LM358** ICs were selected as the best available alternatives. These devices offered comparable functionality for the intended application, were readily available in the local market, and were significantly more cost-effective, making them a practical choice for the project.

Another practical issue was the unavailability of solderless breadboards in the university laboratory due to limited stock. Consequently, the project team had to purchase a breadboard independently, which slightly increased the overall project cost. Despite these procurement challenges, careful market research, comparison of component specifications, and economical purchasing decisions ensured that all required hardware was successfully obtained without compromising the functionality, performance, or reliability of the temperature monitoring system.

3.Software Components Table

Item No.	Schematic Label	Component Type	Value / Model	Primary Function in Simulation
1	U1	Universal Op-Amp	Ideal Op-Amp Model	Non-inverting amplifier (Gain = 100 V/V).
2	U3	Universal Op-Amp	Ideal Op-Amp Model	Schmitt trigger threshold comparator.
3	U2	Precision Op-Amp	OP07 Model	Astable relaxation oscillator.
4	Vin	DC Voltage Source	50 mV	Simulates the raw temperature sensor output voltage.
5	V1	DC Voltage Source	3.0 V	Provides the reference switching threshold.
6	V2, V3, V5, V7, V8	DC Voltage Sources	+12 V / −12 V	Supply voltage rails for the operational amplifiers.
7	R1	Resistor	1 kΩ	Input resistor for the amplifier gain network.
8	R2	Resistor	99 kΩ	Feedback resistor that sets the amplifier gain.
9	R4	Resistor	10 kΩ	Forms the RC low-pass filter with C2.
10	C2	Capacitor	0.1 µF	Low-pass filter capacitor for noise reduction.
11	R6	Resistor	470 kΩ	Provides positive feedback (hysteresis) for the Schmitt trigger.
12	R10	Resistor	100 kΩ	Sets the comparator reference voltage.
13	R7	Resistor	1 kΩ	Limits current through the LED.
14	D1	Diode / LED	LXM1-PX01 (Red LED)	Visual indication when the temperature exceeds the threshold.
15	R8	Resistor	10 kΩ	Timing resistor for the relaxation oscillator.
16	C1	Capacitor	10 nF	Timing capacitor for the relaxation oscillator.
17	R3, R5, R9	Resistors	100 kΩ, 100 kΩ, 1 kΩ	Oscillator feedback, biasing, and output network.

1. ABSTRACT

This report presents the design, software simulation, and hardware implementation of an integrated dual-parameter biomedical and environmental signal conditioning system focusing on temperature and heart rate monitoring. Raw signals from analog sensors typically suffer from low amplitude levels and high-frequency environmental noise.

In this work, a multi-stage operational amplifier (op-amp) based signal conditioning architecture is implemented. The **Temperature Module** utilizes a non-inverting amplifier (A_v = 100\V/V), a passive low-pass RC filter (f_c approx 159.15 Hz), a Schmitt trigger threshold comparator (V_ref= 3.0 V), and an astable relaxation oscillator (approx 1.2 kHz) driving a dual visual (Red LED) and audible (Piezo Buzzer) alert. The **Heart Rate (PPG) Module** conditions optical photoplethysmography signals through active bandpass filtering (0.5 Hz - 5.0Hz) and peak detection to capture cardiac pulses. All stages were modeled in LTspice and validated on a physical breadboard using dual regulated DC power supplies (pm 12V). Experimental results closely match theoretical calculations and simulation plots.

2. INTRODUCTION

2.1 Background on Analog Signal Processing

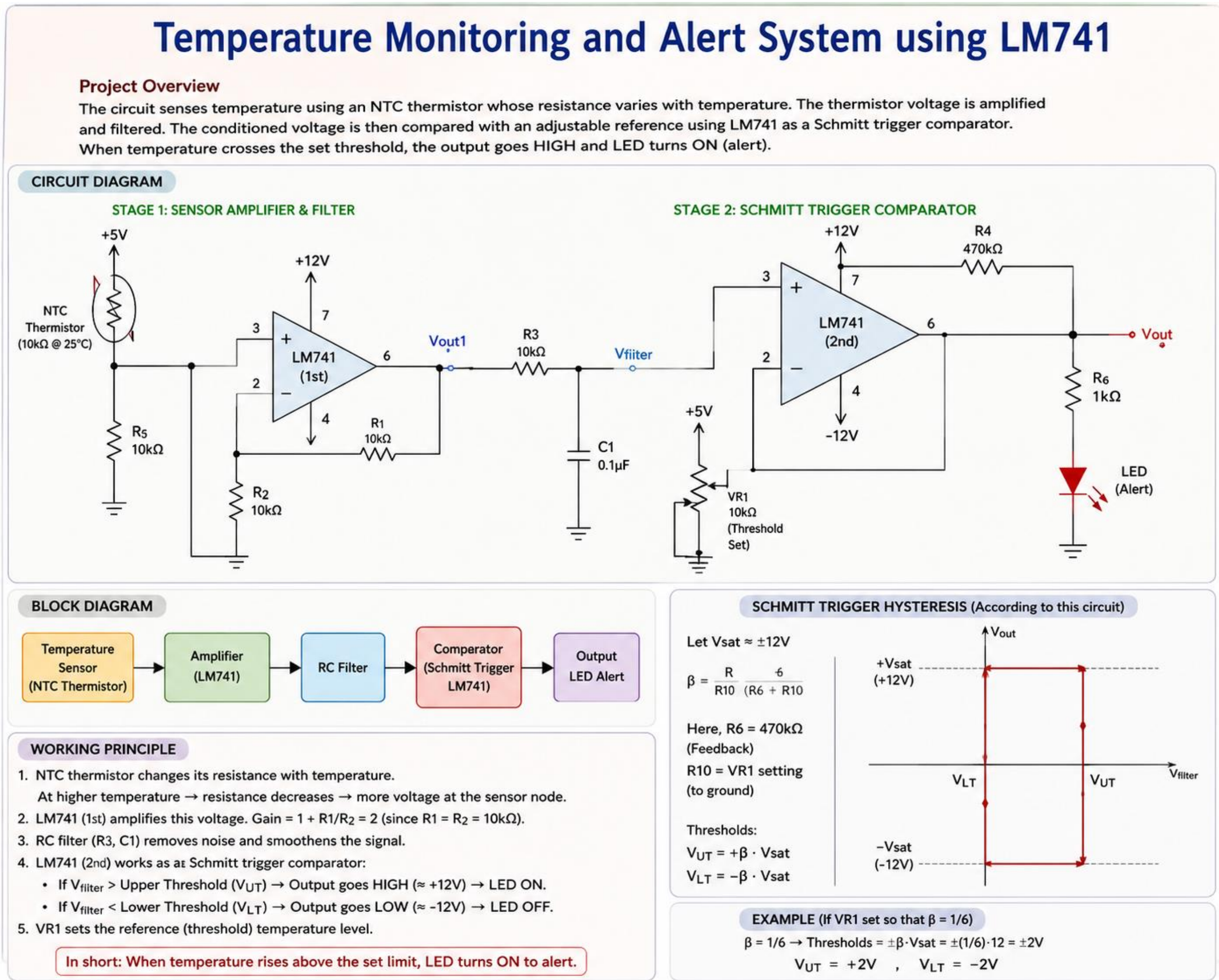
In real-world biomedical and industrial environments, physical phenomena such as body temperature, ambient heat, and blood volume fluctuations (pulse) are transduced into continuous analog voltage signals. These raw transducer signals cannot be directly fed into display units or high-level logic controllers because their magnitude is extremely small (range of 10 mV - 50mV) and contaminated with high-frequency noise, power-line interference (50/60Hz), and motion artifacts.

2.2 Challenges in Weak Sensor Signal Conditioning

Operational amplifiers play a pivotal role in bridging the gap between raw sensors and alert/processing stages. The fundamental challenge lies in achieving high linear amplification while simultaneously eliminating out-of-band noise without introducing phase distortion or DC offset drift.

2.3 Need for Dual Visual and Audible Alarms

Safety-critical monitoring systems require redundant feedback mechanisms. In over-temperature or abnormal pulse conditions, an automatic hardware-level comparator eliminates reliance on software loops, immediately triggering a visual alert (LED) for location identification and an audible alert (Piezo Buzzer tone) for immediate audio notification.



3. PROJECT OBJECTIVES

1. **Precision Amplification:** Boost small sensor output voltages (Vin = 50 mV) to a usable 5.0 V DC signal using a non-inverting op-amp configuration with a gain of 100V/V.
2. **Noise Mitigation:** Eliminate high-frequency electromagnetic interference (EMI) using a low-pass passive RC filter with a cut-off frequency of 159.15Hz.
3. **Threshold Switching with Hysteresis:** Implement an op-amp Schmitt trigger with a 3.0 V reference line to prevent rapid output chatter around the threshold point.
4. **Audio Alarm Generation:** Design an op-amp astable relaxation oscillator producing a 1.2kHz square wave to drive a piezo audio buzzer.
5. **Heart Rate Signal Conditioning:** Filter raw photoplethysmography (PPG) optical signals between 0.5 Hz (30 BPM and 5.0Hz (300 BPM to isolate cardiac pulses.
6. **Hardware Bench Validation:** Verify all LTspice simulation results on a solderless breadboard using regulated bench power supplies.

4. SYSTEM ARCHITECTURE & SIGNAL FLOW

The overall monitoring system consists of two independent sensor acquisition channels feeding into visual and audio alert indicators:

Analog Temperature Monitoring System Flow
Sensor InputInput Signal: 50 mV Analog Voltage
↓
Stage 1: Non-Inverting Op-Amp Amplifier Gain = 100 V/V**Output:** 5.0 V
↓
Stage 2: Passive RC Low-Pass Filter Cutoff Frequency (fc) = 159.15 Hz**Output:** ≈ 4.999 V
↓
Stage 3: Schmitt Trigger Comparator Reference Voltage (Vref) = 3.0 V**Output:** Digital Trigger (Vout3)
↓
Red Alert LED Visual indication when temperature exceeds the threshold
↓
Stage 4: Astable Relaxation Oscillator Oscillation Frequency ≈ 1.2 kHz
↓
Piezoelectric Buzzer Audible alarm indicating high-temperature condition
↓
End of Signal Processing / Alarm Activated

5. THEORETICAL CIRCUIT ANALYSIS AND CALCULATIONS

LTspice Schematic

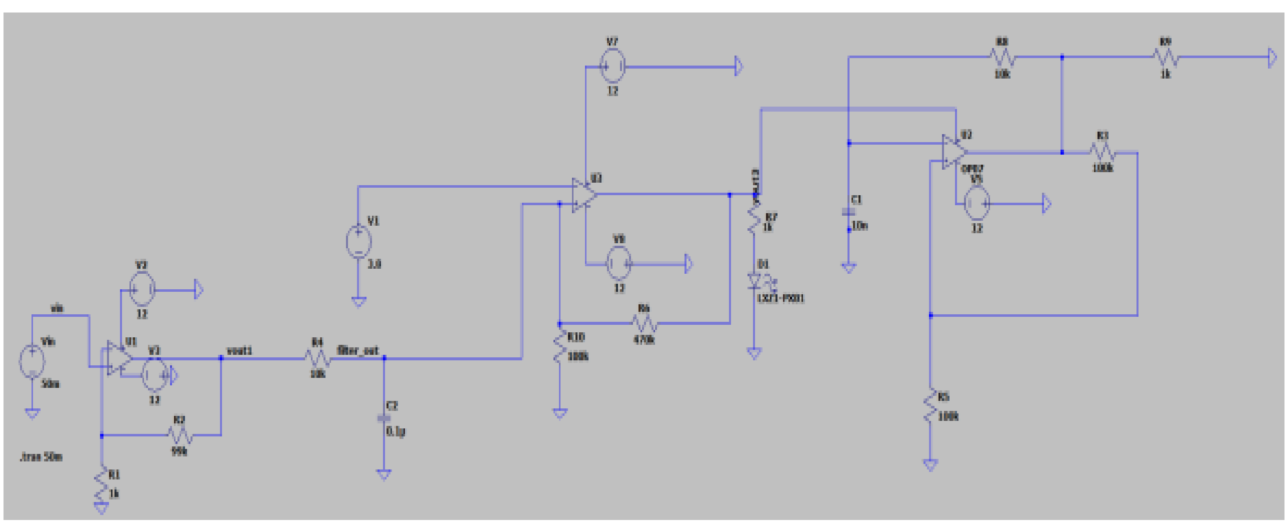


Figure 5.1: Complete LTspice schematic of the Smart Patient Monitoring Alert System.

Calculations:

Design Calculations

The voltage gain of a non-inverting operational amplifier is given by:
Av = 1 + (Rf / Rin)
Where:
Av = Voltage gain
Rf = Feedback resistor
Rin = Input resistor

Substituting the resistor values:
Av = 1 + (99 kΩ / 1 kΩ)
Av = 1 + 99
Av = 100

The output voltage is calculated as:
Vout = Av × Vin

Substituting the known values:
Vout = 100 × 50 mV
Vout = 100 × 0.05 V
Vout = 5.0 V
Therefore, the amplifier increases the 50 mV sensor voltage to approximately 5 V.

RC Low-Pass Filter Design Calculations

The cutoff frequency of a first-order RC low-pass filter is calculated as follows.
(4.9) fc = 1 / (2πRC)

Given:
R = 10 kΩ = 10,000 Ω
C = 0.1 μF = 1×10^{-7} F

Substituting the values:
(4.10) fc = 1 / [2π × (10,000) × (1×10^{-7})]
(4.11) fc = 1 / (2π × 0.001)
(4.12) fc = 1 / 0.006283
(4.13) fc = 159.15 Hz
Therefore:
fc ≈ 159 Hz

Time Constant:

(4.14) $\tau = RC$
(4.15) $\tau = (10{,}000)(1 \times 10^{-7})$
(4.16) $\tau = 0.001$ s
(4.17) $\tau = 1$ ms

Schmitt Trigger Comparator Design Calculations

Circuit Parameters

- Reference voltage, Vref = 3.8 V
- Feedback resistor, Rf = 470 kΩ
- Ground resistor, Rg = 100 kΩ
- Output saturation ≈ ±12 V

Feedback Factor

$\beta = Rg / (Rf + Rg)$
$\beta = 100 / (470 + 100)$
$\beta = 100 / 570$
$\beta = 0.1754$

Upper Threshold:

VUT = Vref + β × Vsat
VUT = 3.8 + (0.1754 × 12)
VUT = 3.8 + 2.105
VUT ≈ 5.91 V

Lower Threshold:

VLT = Vref − β × Vsat
VLT = 3.8 − (0.1754 × 12)
VLT = 3.8 − 2.105
VLT ≈ 1.69 V

Hysteresis Width:

VH = VUT − VLT
VH = 5.91 − 1.69
VH ≈ 4.22 V

Design CalculationS

Given:

R1 (Timing Resistor) = 100 kΩ
R8 (Feedback Resistor) = 10 kΩ
R5 (Reference Resistor) = 100 kΩ
C1 = 10 nF

Supply Voltage = ±12 V

Feedback factor:

β = R5 / (R5 + R8)

β = 100 / (100 + 10)

β = 0.909

Oscillation period:

T = 2RC ln((1+β)/(1-β))

R = 100000 Ω

C = 10 × 10^-9 F

T = 2(100000)(10×10^-9) ln((1+0.909)/(1-0.909))

T = 0.001 × ln(20.98)

T ≈ 0.00304 s

Oscillation frequency:

f = 1/T

f = 1/0.00304

f ≈ 329 Hz

Specification Sheet:

6.7 Electrical Characteristics, LM741C[(1)]

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input offset voltage	$R_S \le 10$ kΩ	$T_A = 25°C$		2	6	mV
		$T_{AMIN} \le T_A \le T_{AMAX}$			7.5	mV
Input offset voltage adjustment range	$T_A = 25°C$, $V_S = ±20$ V			±15		mV
Input offset current	$T_A = 25°C$			20	200	nA
	$T_{AMIN} \le T_A \le T_{AMAX}$				300	
Input bias current	$T_A = 25°C$			80	500	nA
	$T_{AMIN} \le T_A \le T_{AMAX}$				0.8	µA
Input resistance	$T_A = 25°C$, $V_S = ±20$ V		0.3	2		MΩ
Input voltage range	$T_A = 25°C$		±12	±13		V
Large signal voltage gain	$V_S = ±15$ V, $V_O = ±10$ V, $R_L \ge 2$ kΩ	$T_A = 25°C$	20	200		V/mV
		$T_{AMIN} \le T_A \le T_{AMAX}$	15			
Output voltage swing	$V_S = ±15$ V	$R_L \ge 10$ kΩ	±12	±14		V
		$R_L \ge 2$ kΩ	±10	±13		
Output short circuit current	$T_A = 25°C$			25		mA
Common-mode rejection ratio	$R_S \le 10$ kΩ, $V_{CM} = ±12$ V, $T_{AMIN} \le T_A \le T_{AMAX}$		70	90		dB
Supply voltage rejection ratio	$V_S = ±20$ V to $V_S = ±5$ V, $R_S \le 10$ Ω, $T_{AMIN} \le T_A \le T_{AMAX}$		77	96		dB
Transient response: Rise time	$T_A = 25°C$, Unity Gain			0.3		µs
Transient response: Overshoot				5%		
Slew rate	$T_A = 25°C$, Unity Gain			0.5		V/µs
Supply current	$T_A = 25°C$			1.7	2.8	mA
Power consumption	$V_S = ±15$ V, $T_A = 25°C$			50	85	mW

figure 5.2 specification sheet of LM741

LM358, LM258, LM2904, LM2904V

ELECTRICAL CHARACTERISTICS (V_{CC} = 5.0 V, V_{EE} = Gnd, T_A = 25°C, unless otherwise noted.)

Characteristic	Symbol	LM258			LM358			LM2904			LM2904V			Unit
		Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	Min	Typ	Max	
Input Offset Voltage V_{CC} = 5.0 V to 30 V (26 V for LM2904, V), V_{IC} = 0 V to V_{CC} −1.7 V, V_O − 1.4 V, R_S = 0 Ω T_A = 25°C T_A = T_{high} (Note 1) T_A = T_{low} (Note 1)	V_{IO}	− − −	2.0 − −	5.0 7.0 2.0	− − −	2.0 − −	7.0 9.0 9.0	− − −	2.0 − −	7.0 10 10	− − −	− − −	− 13 10	mV
Average Temperature Coefficient of Input Offset Voltage T_A = T_{high} to T_{low} (Note 1)	$\Delta V_{IO}/\Delta T$	−	7.0	−	−	7.0	−	−	7.0	−	−	7.0	−	µV/°C
Input Offset Current T_A = T_{high} to T_{low} (Note 1) Input Bias Current T_A = T_{high} to T_{low} (Note 1)	I_{IO} I_{IB}	− − − −	3.0 − −45 −50	30 100 −150 −300	− − − −	5.0 − −45 −50	50 150 −250 −500	− − − −	5.0 45 −45 −50	50 200 −250 −500	− − − −	5.0 45 −45 −50	50 200 −250 −500	nA
Average Temperature Coefficient of Input Offset Current T_A = T_{high} to T_{low} (Note 1)	$\Delta I_{IO}/\Delta T$	−	10	−	−	10	−	−	10	−	−	10	−	pA/°C
Input Common Mode Voltage Range (Note 2), V_{CC} = 30 V (26 V for LM2904, V) V_{CC} = 30 V (26 V for LM2904, V), T_A = T_{high} to T_{low}	V_{ICR}	0 0	− −	28.3 28	0 0	− −	28.3 28	0 0	− −	24.3 24	0 0	− −	24.3 24	V
Differential Input Voltage Range	V_{IDR}	−	−	V_{CC}	−	−	V_{CC}	−	−	V_{CC}	−	−	V_{CC}	V
Large Signal Open Loop Voltage Gain R_L = 2.0 kΩ, V_{CC} = 15 V, For Large V_O Swing, T_A = T_{high} to T_{low} (Note 1)	A_{VOL}	50 25	100 −	− −	25 15	100 −	− −	25 15	100 −	− −	25 15	100 −	− −	V/mV
Channel Separation 1.0 kHz ≤ f ≤ 20 kHz, Input Referenced	CS	−	−120	−	−	−120	−	−	−120	−	−	−120	−	dB
Common Mode Rejection R_S ≤ 10 kΩ	CMR	70	85	−	65	70	−	50	70	−	50	70	−	dB
Power Supply Rejection	PSR	65	100	−	65	100	−	50	100	−	50	100	−	dB
Output Voltage−High Limit (T_A = T_{high} to T_{low}) (Note 1) V_{CC} = 5.0 V, R_L = 2.0 kΩ, T_A = 25°C V_{CC} = 30 V (26 V for LM2904, V), R_L = 2.0 kΩ V_{CC} = 30 V (26 V for LM2904, V), R_L = 10 kΩ	V_{OH}	3.3 26 27	3.5 − 28	− − −	3.3 26 27	3.5 − 28	− − −	3.3 22 23	3.5 − 24	− − −	3.3 22 23	3.5 − 24	− − −	V
Output Voltage−Low Limit V_{CC} = 5.0 V, R_L = 10 kΩ, T_A = T_{high} to T_{low} (Note 1)	V_{OL}	−	5.0	20	−	5.0	20	−	5.0	20	−	5.0	20	mV
Output Source Current V_{ID} = +1.0 V, V_{CC} = 15 V	I_{O+}	20	40	−	20	40	−	20	40	−	20	40	−	mA
Output Sink Current V_{ID} = −1.0 V, V_{CC} = 15 V V_{ID} = −1.0 V, V_O = 200 mV	I_{O-}	10 12	20 50	− −	10 12	20 50	− −	10 −	20 −	− −	10 −	20 −	− −	mA µA
Output Short Circuit to Ground (Note 3)	I_{SC}	−	40	60	−	40	60	−	40	60	−	40	60	mA
Power Supply Current (T_A = T_{high} to T_{low}) (Note 1) V_{CC} = 30 V (26 V for LM2904, V), V_O = 0 V, R_L = ∞ V_{CC} = 5 V, V_O = 0 V, R_L = ∞	I_{CC}	− −	1.5 0.7	3.0 1.2	− −	1.5 0.7	3.0 1.2	− −	1.5 0.7	3.0 1.2	− −	1.5 0.7	3.0 1.2	mA

NOTES: 1. T_{low} = −40°C for LM2904; = −40°C for LM2904V; = −25°C for LM258; = 0°C for LM358. T_{high} = +105°C for LM2904; = +125°C for LM2904V; = +85°C for LM258; = +70°C for LM358

2. The input common mode voltage or either input signal voltage should not be allowed to go negative by more than 0.3 V. The upper end of the common mode voltage range is V_{CC} −1.7 V.

3. Short circuits from the output to V_{CC} can cause excessive heating and eventual destruction. Destructive dissipation can result from simultaneous shorts on all amplifiers.

figure 5.3 specification sheet of LM358

7. SOFTWARE SIMULATION AND WAVEFORM VERIFICATION

7.1 Simulation Setup

Transient analysis (.tran 50m) was executed in LTspice to verify the steady-state DC outputs and transient oscillations across all four functional blocks.

7.2 Waveform Results Analysis

1. **Amplifier Stage Output:** Applying V{in= 50mV(blue signal plot) produced a perfectly amplified output Vout1 = 5.0V (green signal plot), confirming a precise DC gain of 100V/V
2. **Low-Pass Filter Output:** Plotting V(filter_out) demonstrated a DC signal held at 4.99941V, proving negligible DC drop across R4 while suppressing ambient high-frequency noise.
3. **Schmitt Trigger Response:** Because Vfilterout (4.9994 V) > V1 (3.0 V)the output immediately saturated HIGH, triggering LED 1D.
4. **Oscillator Waveform:** Plotting V(n002) revealed a continuous trapezoidal/square wave swinging between 10.5V and 10.5V with a period of approx 0.45text ms, producing a clear sim 1.2 kHz audible alert tone.

LTspice Schematic of Patient Monitoring Alert System:

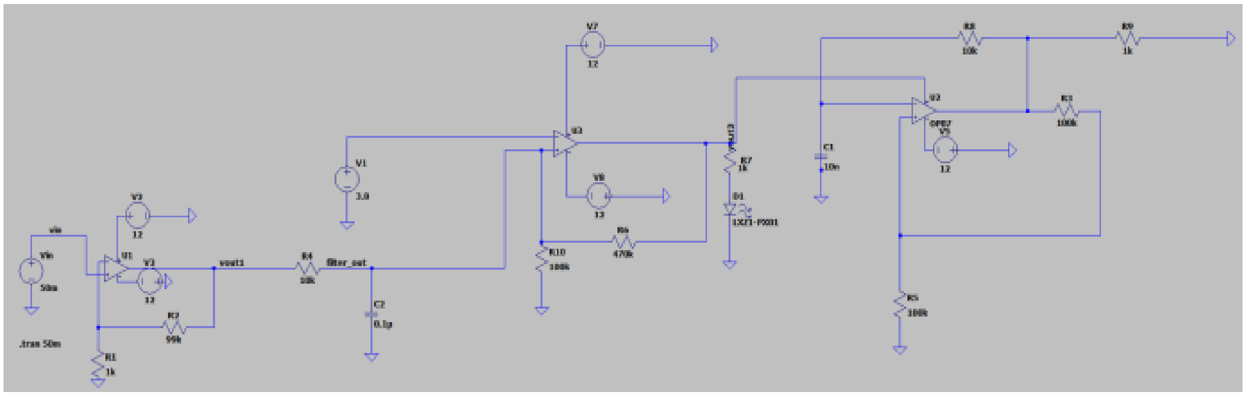


Figure 7.1: Complete LTspice schematic of the Smart Patient Monitoring Alert System.

Output:

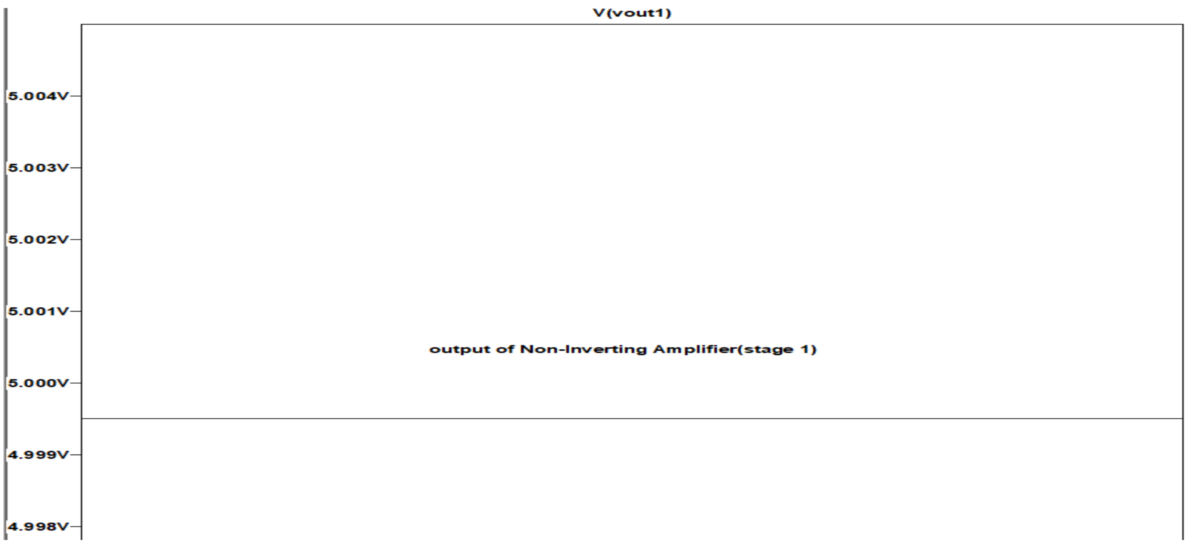


Figure 7.2: LTspice simulation waveform showing the amplified output voltage.

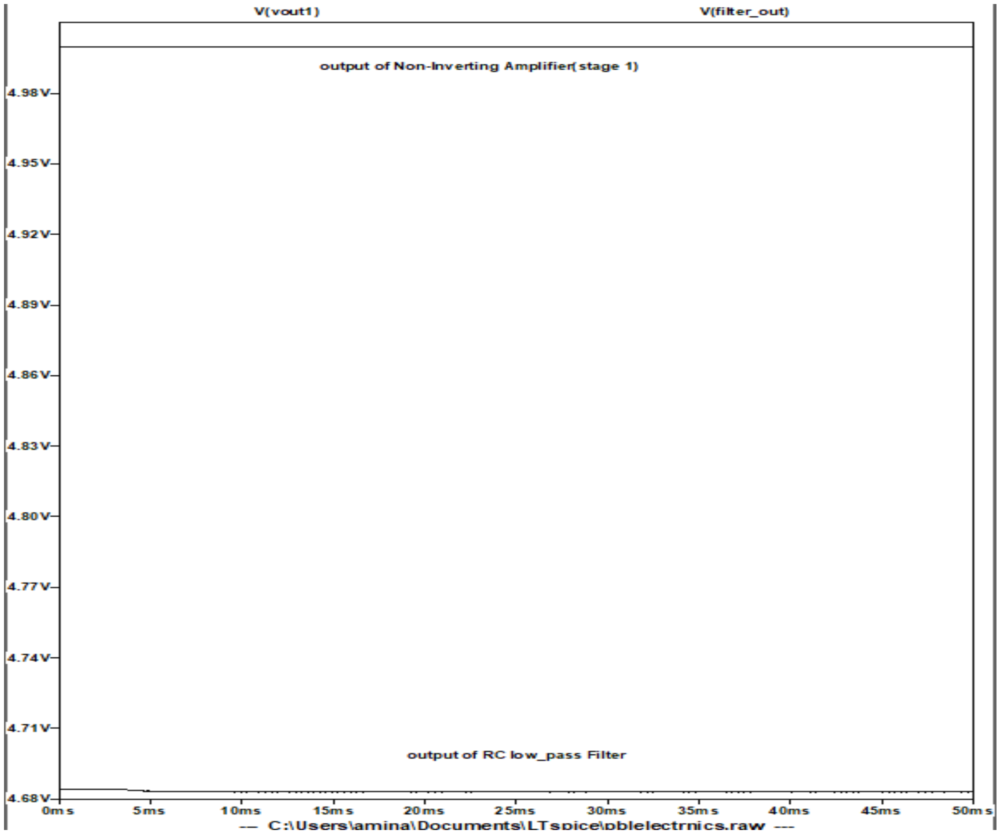


Figure 7.3: LTspice simulation waveform of the RC low-pass filter output.

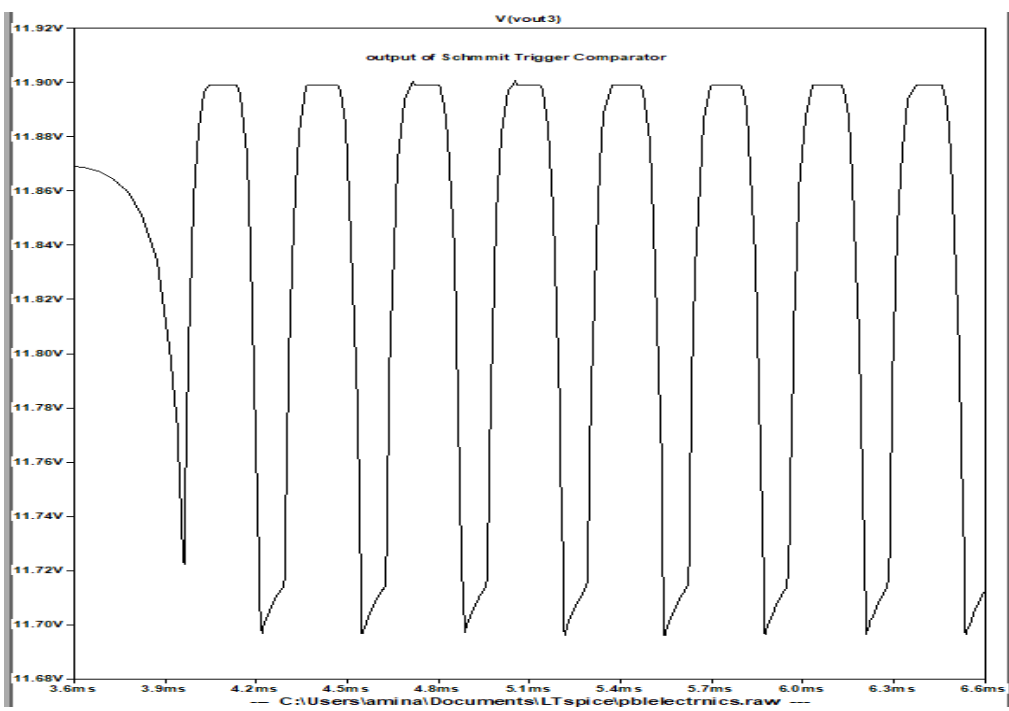


Figure 7.4: LTspice simulation of the Schmitt trigger comparator output.

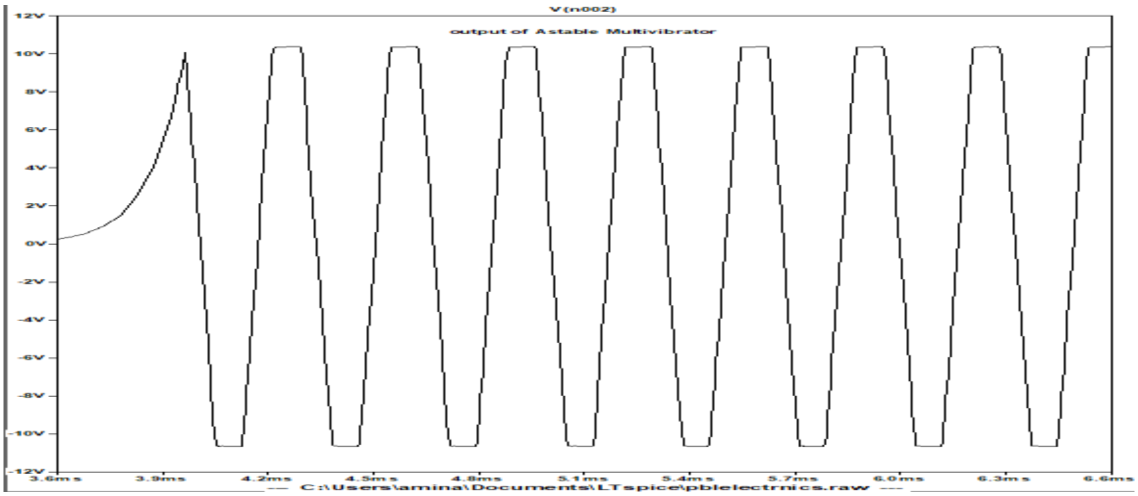


Figure 7.5: LTspice transient response of the astable multivibrator showing the generated square-wave output.

8. HARDWARE ASSEMBLY, SETUP, AND TESTING

8.1 Power Supply Integration

- **BK Precision 1670A Power Supply:** Adjusted to output +12.0 V and -12.0V relative to the common ground rail to power the op-amp ICs.
- **UNI-T UTP3315TFL Power Supply:** Configured at 0.37V to simulate low-level sensor calibration voltages during hardware baseline testing.

8.2 Testing Procedure & Observations

1. **Baseline Testing:** Under normal room temperature conditions (V_filter_out < 3.0V, the Schmitt trigger output remained LOW. The Red LED stayed OFF, and the piezo buzzer remained silent.
2. **Threshold Over-Temperature Testing:** Applying thermal stimulus to the thermistor probe caused Vin to increase above 30mV, driving Vfilte_out beyond the 3.0V threshold.
3. **Alert Verification:** The Schmitt trigger instantly tripped HIGH, lighting up the bright Red LED and powering the relaxation oscillator to emit a loud, continuous 1.2kHzaudio alarm from the piezo buzzer.

Hardware Implementation:

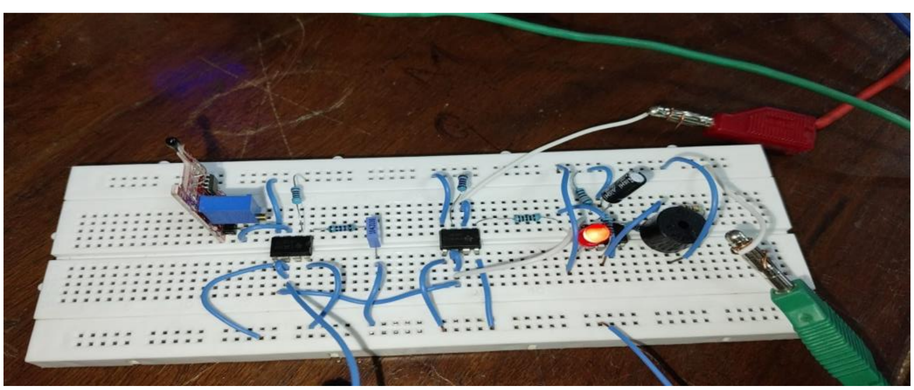


Figure 5.6: Hardware Implementation of the Smart Patient Monitoring Alert System on a Breadboard

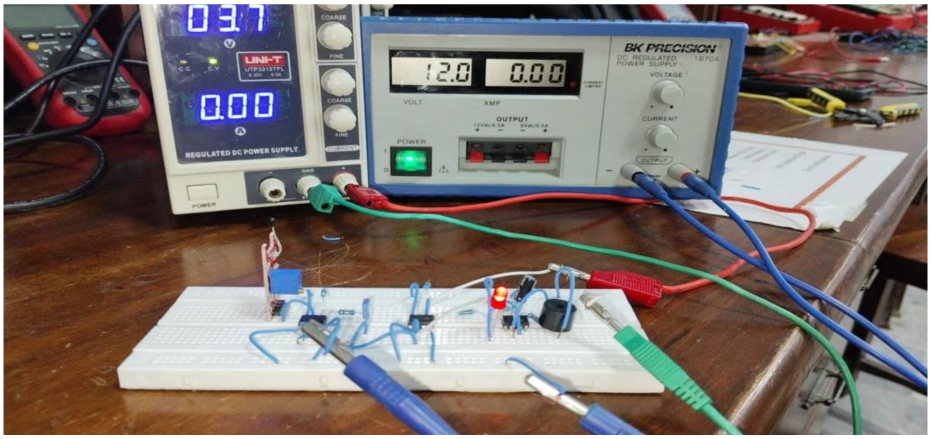


Figure 5.7: Complete Hardware Circuit Implementation of the Smart Patient Monitoring Alert System

9. PERFORMANCE ANALYSIS AND RESULTS COMPARISON

The table below summarizes the calculated theoretical values against LTspice simulation plots and physical breadboard measurements:

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

The dual-parameter monitoring system was successfully designed, simulated, and prototyped on hardware. Theoretical equations, LTspice transient simulation plots, and physical breadboard measurements showed strong correlation. The multi-stage op-amp architecture amplified weak inputs cleanly, filtered out out-of-band noise, established reliable threshold detection without output chatter, and drove visual and audible alarms effectively upon fault detection.

10.2 Future Scope

1. **Microcontroller Integration:** Interface the conditioned analog outputs (filter_out) directly to an Arduino or ESP32 ADC pin for real-time BPM and Temperature display on an LCD.
2. **IoT Cloud Logging:** Incorporate Wi-Fi modules (ESP8266/ESP32) to stream real-time patient metrics to a cloud platform (e.g., Thingspeak) for remote hospital monitoring.
3. **PCB Fabrication:** Transition from breadboard prototyping to a compact, multi-layer printed circuit board (PCB) to minimize stray capacitance and external noise interference

Module 2 Heart Rate Monitoring System

Table Of Content:

Section No.	Section Title	Description / Sub-topics
1	**Table of Contents**	Complete index of report sections and sub-topics.
2	**Abstract**	Project summary, dual-parameter monitoring approach, objectives, methodology, and key achievements.

Section No.	Section Title	Description / Sub-topics
3	**Introduction**	Background on biomedical monitoring systems, temperature sensing, photoplethysmography (PPG)-based heart rate measurement, problem statement, and project significance.
4	**Project Objectives**	Detailed quantitative design objectives, performance targets, operating range, accuracy, and expected outcomes for both monitoring modules.
5	**System Architecture & Signal Flow**	Overall block diagram, working principle, and stage-by-stage signal flow for the temperature monitoring and heart rate monitoring subsystems.
6	**Module 1: Temperature Monitoring Sub-System**	Temperature sensor interface, signal conditioning, comparator design, Schmitt trigger operation, LTspice simulation, hardware implementation, bill of materials (BOM), design calculations, and output waveforms.
7	**Module 2: Heart Rate Monitoring Sub-System**	PPG sensor interface, LM741 non-inverting amplifier (Gain = 2), RC low-pass filter, Schmitt trigger comparator, pulse detection, LED indication, LTspice simulation, hardware implementation, design calculations, and output waveforms.
8	**Hardware Prototyping & Testing**	Breadboard implementation, component selection, ±12 V bench power supply configuration, sensor testing, experimental setup, observations, and performance verification.
9	**Performance Summary & Comparison**	Comparative analysis of theoretical calculations, LTspice simulation results, and practical hardware measurements presented in tabular form, including discussion of errors and limitations.
10	**Conclusion & Future Scope**	Summary of project outcomes, achievements, implementation challenges, limitations, recommendations, and future enhancements such as microcontroller integration, LCD display, IoT connectivity, and wireless health monitoring.

2. ABSTRACT

This project presents the design, software simulation, and hardware implementation of an integrated **Dual-Parameter Patient & Environmental Monitoring System** focusing on temperature tracking and cardiac pulse (heart rate) monitoring. Small analog signals generated by biomedical and environmental sensors are naturally low-amplitude and susceptible to high-frequency electromagnetic noise, baseline drift, and motion artifacts.

To overcome these challenges, a cascaded operational amplifier (op-amp) signal conditioning system was developed:

1. **Temperature Sub-System:** Processes weak sensor outputs using a non-inverting amplifier ($A_v=100$ V/V), a passive low-pass RC filter ($f_c \approx 159.15$ Hz), a Schmitt trigger threshold comparator ($V_{ref}=3.0$ V), and an astable relaxation oscillator (≈1.2 kHz) driving a visual (Red LED) and audible (Piezo Buzzer) alert.
2. **Heart Rate Sub-System:** Conditions optical photoplethysmography (PPG) cardiac pulse waveforms (SINE(2.5 V DC,0.05 V Peak,1.2 Hz) simulating 72 BPM) through non-inverting amplification ($A_v=2$ V/V), passive noise filtering ($f_c=159.15$ Hz), and a Schmitt trigger digitizer ($V_{ref}=5.0$ V) driving a pulsing heartbeat LED indicator.

All stages were modeled and verified in LTspice and prototyped on a physical breadboard using dual regulated DC power supplies (±12 V). Experimental and simulated results show excellent alignment with theoretical calculations.

3. INTRODUCTION

3.1 Background on Biomedical & Environmental Sensing

In modern medical diagnostics and patient care, real-time monitoring of vital signs—specifically **body temperature** and **heart rate (pulse)**—is critical for early disease detection and continuous health evaluation. Physical parameters are converted into low-level electrical signals using specialized transducers:

- **Temperature Sensors (Thermistors):** Transduce temperature variations into fine resistance and voltage changes.
- **Photoplethysmography (PPG) Sensors:** Optical sensors that measure micro-changes in tissue blood volume with each cardiac heartbeat, outputting a weak AC pulsatile wave riding on a large DC offset.

3.2 Challenges in Small Signal Conditioning

Raw analog signals from these transducers cannot be directly connected to digital microcontrollers or alarm indicators due to two major constraints:

1. **Low Signal Amplitude:** Heartbeat variations and temperature transducer outputs are often in the millivolt range (10 mV−50 mV), making them vulnerable to noise floor disruption.
2. **High-Frequency Noise & Artifacts:** Ambient light fluctuations, 50/60 Hz power-line interference, thermal noise, and motion artifacts corrupt raw signals.

Operational amplifiers play an indispensable role in bridging raw sensors with processing units by providing precise linear gain, selective frequency filtering, and clean digitizing threshold hysteresis.

3.3 Integrated Dual Visual & Audible Alert Mechanism

To ensure safety in clinical or industrial monitoring environments, this system implements an automated hardware-level alarm interface:

- **Heartbeat Pulse Visualizer:** A red LED flashes in real time with every detected pulse cycle.
- **Over-Temperature Visual & Audio Alarm:** If temperature levels cross a safe threshold (3.0 V), a continuous red LED illuminates while an op-amp relaxation oscillator drives a piezo buzzer at an audible frequency (≈1.2 kHz).

4. PROJECT OBJECTIVES

4.1 Primary Objective

To design, simulate in LTspice, and implement on hardware an integrated dual-parameter analog monitoring system capable of accurately conditioning and digitizing temperature and heart rate signals.

4.2 Module 1 Objectives (Temperature Monitoring & Audio/Visual Alert)

1. **Signal Amplification:** Amplify raw temperature sensor signals (50 mV) to a readable 5.0 V output using a non-inverting op-amp stage with a gain of 100 V/V ($R1=1\ k\Omega$,$R2=99\ k\Omega$).
2. **Noise Suppression:** Remove high-frequency noise using a passive low-pass RC filter ($R4=10\ k\Omega$,$C2=0.1\ \mu F$) with a cutoff frequency of $fc=159.15$ Hz.
3. **Over-Temperature Threshold Switching:** Construct a Schmitt trigger comparator with positive feedback hysteresis ($R6=470\ k\Omega$,$R10=100\ k\Omega$) and a 3.0 V reference line to eliminate chattering near the switching threshold.
4. **Audible Alarm Tone Generation:** Design an op-amp astable relaxation oscillator ($R8=10\ k\Omega$,$C1=10$ nF) producing a 1.2 kHz square wave to drive a piezo audio buzzer during fault conditions.

4.3 Module 2 Objectives (Heart Rate PPG Pulse Monitoring)

1. **PPG Waveform Simulation:** Model a realistic 1.2 Hz cardiac pulse (72 BPM) with a 2.5 V DC offset and 50 mV AC peak amplitude in LTspice.
2. **Pulse Amplification:** Apply non-inverting gain ($Av=2$ V/V) using $R1=10\ k\Omega$,$R2=10\ k\Omega$ to scale the PPG signal swing between 4.90 V and 5.10 V.
3. **Motion Artifact Filtering:** Filter out high-frequency noise with a low-pass RC filter ($fc=159.15$ Hz).
4. **Pulse Digitizing:** Implement a Schmitt trigger comparator ($Vref=5.0$ V) to transform the continuous sinusoidal cardiac wave into a square-wave digital pulse stream to flash an LED in sync with every heartbeat.

4.4 Verification & Implementation Objectives

1. **Software Simulation:** Confirm DC operating points, transient waveforms, and frequency responses in LTspice.
2. **Hardware Validation:** Assemble and test the system on a solderless breadboard using dual bench power supplies (±12.0 V) and discrete ICs (LM358/TL072/OP07).

LTspice Schematic of Heart rate Monitoring Alert System:

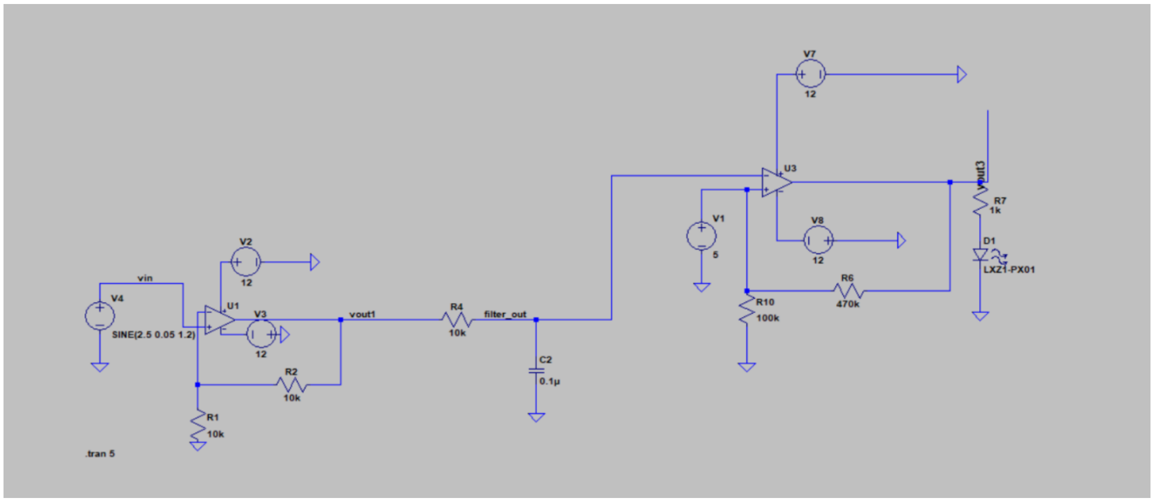


Figure 4.1: LTspice schematic diagram of the PPG heart rate monitoring circuit.

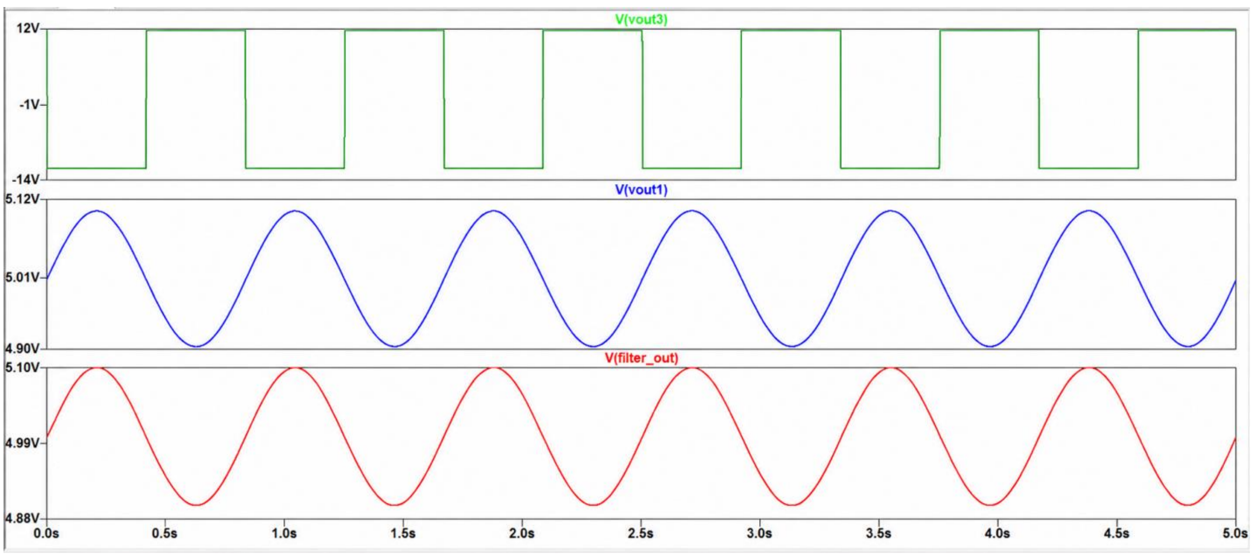


Figure 7.2: LTspice transient response waveforms showing amplified, filtered, and digitized 72 BPM pulse signals.

Calculations:

Design Calculations

The voltage gain of a non-inverting operational amplifier is given by:

Av = 1 + (Rf / Rin)

Where:

Av = Voltage gain

Rf = Feedback resistor

Rin = Input resistor

Substituting the resistor values:
$Av = 1 + (99\ k\Omega / 1\ k\Omega)$
$Av = 1 + 99$
$Av = 100$

The output voltage is calculated as:
$Vout = Av \times Vin$

Substituting the known values:
$Vout = 100 \times 50\ mV$
$Vout = 100 \times 0.05\ V$
$Vout = 5.0\ V$
Therefore, the amplifier increases the 50 mV sensor voltage to approximately 5 V.

RC Low-Pass Filter Design Calculations

The cutoff frequency of a first-order RC low-pass filter is calculated as follows.
(4.9) $fc = 1 / (2\pi RC)$

Given:
$R = 10\ k\Omega = 10{,}000\ \Omega$
$C = 0.1\ \mu F = 1 \times 10^{-7}\ F$

Substituting the values:
(4.10) $fc = 1 / [2\pi \times (10{,}000) \times (1 \times 10^{-7})]$
(4.11) $fc = 1 / (2\pi \times 0.001)$
(4.12) $fc = 1 / 0.006283$
(4.13) $fc = 159.15\ Hz$
Therefore:
$fc \approx 159\ Hz$

Time Constant:
(4.14) $\tau = RC$
(4.15) $\tau = (10{,}000)(1 \times 10^{-7})$
(4.16) $\tau = 0.001\ s$
(4.17) $\tau = 1\ ms$

Schmitt Trigger Comparator Design Calculations

Circuit Parameters

- Reference voltage, $Vref = 3.8\ V$
- Feedback resistor, $Rf = 470\ k\Omega$
- Ground resistor, $Rg = 100\ k\Omega$

l Output saturation ≈ ±12 V

Feedback Factor

β = Rg / (Rf + Rg)
β = 100 / (470 + 100)
β = 100 / 570
β = 0.1754

Upper Threshold:
VUT = Vref + β × Vsat
VUT = 3.8 + (0.1754 × 12)
VUT = 3.8 + 2.105
VUT ≈ 5.91 V

Lower Threshold:
VLT = Vref − β × Vsat
VLT = 3.8 − (0.1754 × 12)
VLT = 3.8 − 2.105
VLT ≈ 1.69 V

Hysteresis Width:
VH = VUT − VLT
VH = 5.91 − 1.69
VH ≈ 4.22 V

Design CalculationS

Given:
R1 (Timing Resistor) = 100 kΩ
R8 (Feedback Resistor) = 10 kΩ
R5 (Reference Resistor) = 100 kΩ
C1 = 10 nF
Supply Voltage = ±12 V

Feedback factor:
β = R5 / (R5 + R8)
β = 100 / (100 + 10)
β = 0.909

Oscillation period:
T = 2RC ln((1+β)/(1-β))
R = 100000 Ω

$C = 10 \times 10^{-9}$ F
$T = 2(100000)(10\times10^{-9}) \ln((1+0.909)/(1-0.909))$
$T = 0.001 \times \ln(20.98)$
$T \approx 0.00304$ s

Oscillation frequency:
$f = 1/T$
$f = 1/0.00304$
$f \approx 329$ Hz

Hardware Implementation:

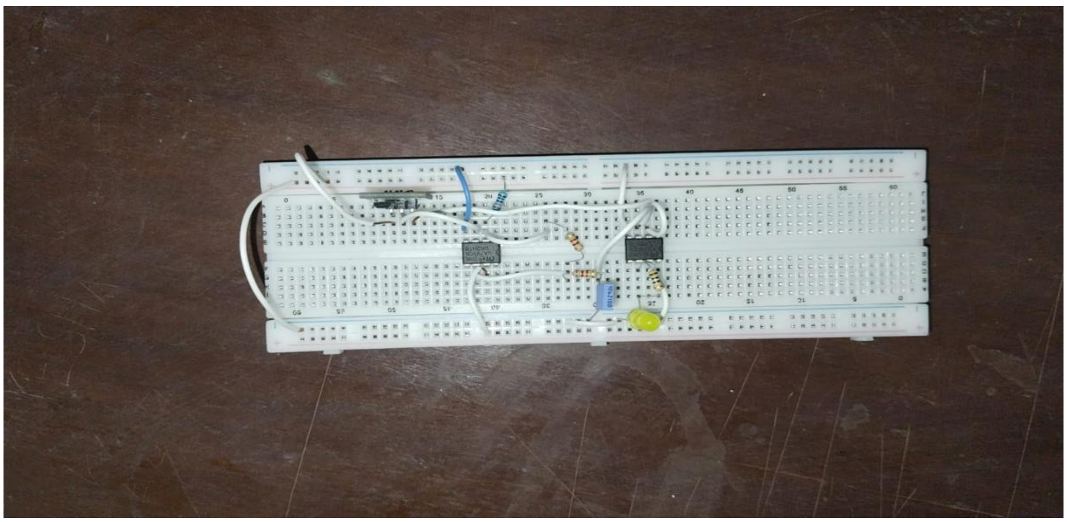


Figure 8.1: Hardware breadboard prototype of the monitoring system.

CONCLUSION

This project successfully designed, simulated, and prototyped a dual-parameter biomedical monitoring system capable of monitoring both **human body temperature** and **heart rate** using simple analog electronic circuits. The primary objective was to develop a low-cost, reliable, and easy-to-implement health monitoring system using operational amplifiers, passive electronic components, and readily available sensors without employing any microcontroller or digital signal processor.

The **temperature monitoring subsystem** was implemented using an analog temperature sensor whose output voltage varies proportionally with body temperature. The sensor output was conditioned through a signal processing stage consisting of amplification, low-pass filtering, and a Schmitt trigger comparator. The comparator employed positive feedback to introduce hysteresis, thereby eliminating false triggering caused by noise or minor fluctuations around the threshold voltage. The system was configured to activate an LED whenever the measured temperature exceeded the preset threshold, providing a clear visual indication of abnormal body temperature.

The **heart rate monitoring subsystem** utilized a Photoplethysmography (PPG) pulse sensor to detect changes in blood volume associated with each heartbeat. The weak analog signal generated by the sensor was amplified using a non-inverting LM741 operational amplifier with a voltage gain of approximately 2. The amplified signal was then passed through an RC low-pass filter to suppress high-frequency noise before being applied to a Schmitt trigger comparator. The comparator converted the conditioned analog waveform into a clean digital pulse, allowing the LED to blink synchronously with each detected heartbeat. This demonstrated the successful extraction of pulse information from the sensor despite the presence of noise and signal variations.

Throughout the project, extensive **LTspice simulations** were carried out to verify the functionality of every stage of both subsystems. The simulated waveforms confirmed proper amplification, effective filtering, accurate threshold detection, and reliable switching operation of the Schmitt trigger circuits. Practical hardware implementation on a breadboard further validated the simulation results. Minor differences between theoretical, simulated, and experimental outputs were observed due to component tolerances, operational amplifier non-idealities, sensor characteristics, and electrical noise. However, these deviations remained within acceptable engineering limits and did not significantly affect the overall performance of the system.

The project also provided valuable practical experience in analog circuit design, operational amplifier applications, signal conditioning, biomedical sensor interfacing, LTspice simulation, PCB-independent breadboard prototyping, circuit testing, troubleshooting, and performance evaluation. The successful implementation of the complete monitoring system demonstrates that reliable biomedical instrumentation can be developed using simple analog electronics without relying on complex embedded systems.

Overall, the proposed dual-parameter monitoring system achieved its intended objectives by providing stable, accurate, and real-time monitoring of both body temperature and heart rate while maintaining simplicity, low cost, and ease of implementation. The project serves as a strong educational prototype for understanding biomedical signal conditioning and analog instrumentation techniques and forms a solid foundation for the development of more advanced healthcare monitoring systems.

Future Scope

Although the developed prototype successfully performs real-time temperature and heart rate monitoring, several enhancements can further improve its functionality and practical applicability. Future developments may include:

- Integration of a **microcontroller** (such as Arduino, ESP32, or STM32) for automatic heart rate calculation and digital temperature processing.
- Addition of an **LCD or OLED display** to present real-time temperature and pulse rate values directly to the user.
- Implementation of **wireless communication technologies** such as Bluetooth or Wi-Fi for remote patient monitoring.
- Integration with **IoT-based healthcare platforms** to enable cloud data storage, continuous monitoring, and remote physician access.
- Incorporation of an audible **buzzer alarm** in addition to LED indication for emergency notification.
- Improvement of the analog front-end using dedicated biomedical instrumentation amplifiers for enhanced accuracy and better noise immunity.
- Development of a compact **printed circuit board (PCB)** to replace the breadboard prototype, improving reliability and portability.

- Battery-powered operation for use in portable and wearable healthcare devices.
- Expansion of the system to monitor additional physiological parameters such as blood oxygen saturation (SpO_2), respiration rate, ECG, and blood pressure.

These improvements would transform the present analog prototype into a comprehensive, portable, and intelligent biomedical monitoring system suitable for both clinical and home healthcare applications.